

To prove:

$$\forall n : \text{nat}, ((n+1) =? 0) = \text{false}$$

$$\Leftrightarrow \forall n : \text{nat}, (\text{eqb } (\text{plus } n \text{ (SO)}) 0) = \text{false}$$

Let n be a natural number

We show that $((n+1) =? 0) = \text{false}$

case analysis on n .

case $n=0$. We show that

$$((0+1) =? 0) = \text{false}$$

$1 = \text{SO}$

$$(\text{SO} =? 0) = \text{false}$$

$\text{false} = \text{false}$

case $n = \text{S } n'$

We show that

$$((\text{S } n') + 1 =? 0) = \text{false}$$

$\rightarrow \text{S } (n' + 1)$
 false

```
Fixpoint eqb (n m : nat) : bool :=
  match n with
  | 0 => match m with
        | 0 => true
        | S m' => false
        end
  | S n' => match m with
            | 0 => false
            | S m' => eqb n' m'
            end
  end.
```

```
Fixpoint plus (n : nat) (m : nat) : nat :=
  match n with
  | 0 => m
  | S n' => S (plus n' m)
  end.
```

By the definition of "+" and "=" $((0+1) =? 0)$ evaluates to false.

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To prove: $\forall n:\text{nat}, \overbrace{n+0=n}^{P(n)=n+0=n}$

Let n be a natural number.

Proof by induction

Case analysis on n .

case $n=0$. We show that $\underbrace{0+0}_{0}=0$ By definition of "+" $0+0$ evaluates to 0.

case $n=S n'$ We show that $\underbrace{S n' + 0}_{S(n'+0)} = S n'$

" $P(n') \Rightarrow P(S n')$ "

where $n'+0=n'$ (IH n') we are left to show $S(n'+0) = S n'$

it would be sufficient to know $\boxed{n'+0=n'}$ $\swarrow$

By the inductive hypothesis (IH n') this is equivalent to $S n' = S n'$ $\square$

```
Fixpoint plus (n : nat) (m : nat) : nat :=
  match n with
  | 0 => m
  | S n' => S (plus n' m)
  end.
```